

Solução do Problema BNL2 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: BNL2
- Número de Variáveis: 3489
- Número de Restrições: 2324
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 1.116e-14
- Valor Primal: 1.811e+03
- Valor Dual: 1.811e+03
- Gap: 3.118e-19
- Número de Iterações: 32

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z)
com valores ≥ 0 do problema BNL2

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	4.193e+01	9	2.229e-01
2	4.193e+01	10	7.995e-01
3	3.307e+01	17	4.400e-01
4	3.307e+01	18	2.351e+00
5	1.250e+02	25	4.400e-01
8	1.250e+02	26	4.804e+01
11	1.250e+02	31	1.133e-03
12	1.250e+02	34	1.600e-02
13	9.451e+01	39	1.600e-02
16	9.451e+01	40	4.400e-02
19	9.451e+01	48	6.661e-16
20	9.451e+01	49	4.656e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
21	7.000e+01	50	1.170e+00
24	7.000e+01	52	7.638e+00
27	7.000e+01	54	2.626e+00
28	7.000e+01	55	6.661e-16
32	6.481e+01	56	6.661e-16
33	6.481e+01	57	2.200e-01
35	6.481e+01	59	7.772e-16
38	6.481e+01	61	7.772e-16
41	1.240e+02	62	4.656e+00
42	1.092e+02	65	7.638e+00
45	9.317e+00	67	2.626e+00
46	3.634e+00	70	2.200e-01
47	3.605e-01	76	4.656e+00
58	5.101e+00	77	1.410e+00
60	8.185e-02	79	7.638e+00
68	1.399e-01	81	2.626e+00
71	4.827e-01	84	2.200e-01
72	6.874e-01	86	4.441e-16
74	6.734e-01	89	4.656e+00
85	3.475e+00	90	1.300e+00
87	7.065e-02	92	7.638e+00
88	1.625e-01	94	2.626e+00
96	1.364e-01	95	4.441e-16
98	1.335e-01	97	2.200e-01
99	5.020e-01	100	4.441e-16
104	2.604e+00	101	4.441e-16
106	2.083e+00	102	4.656e+00
115	1.106e-01	103	6.912e+00
116	4.379e+01	105	7.638e+00
117	2.260e+00	107	2.626e+00
120	1.300e+01	108	4.441e-16
121	9.364e+00	109	3.331e-16
128	1.714e+01	110	2.200e-01
129	4.650e+01	112	1.474e+00
130	1.941e+01	113	5.669e+01
131	2.164e+01	118	6.973e-01
132	5.020e+01	119	7.132e-02
133	1.845e+01	123	3.331e-16
138	2.667e+01	125	6.973e-01
142	9.561e+01	126	7.132e-02
143	5.367e+01	127	2.220e-16
144	7.539e+01	134	6.551e-02
145	4.233e+01	135	4.335e-01
146	2.000e+02	136	4.951e-02
147	5.000e+01	137	3.330e-01
149	7.500e+01	139	3.715e-01
150	5.000e+01	140	2.763e-02
152	2.500e+02	141	6.608e-02
153	1.250e+02	151	3.326e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
154	1.870e+02	158	3.761e-01
157	9.252e+01	159	1.520e+00
160	1.870e+02	166	3.761e-01
161	9.252e+01	167	2.925e+01
162	1.589e+02	174	3.047e-03
165	8.890e+01	175	3.354e-05
168	1.589e+02	178	1.210e-02
169	8.890e+01	183	7.300e-03
172	2.845e+01	184	3.340e-02
176	1.639e+02	192	1.332e-15
177	9.910e+01	193	8.559e+00
179	1.639e+02	194	7.096e-01
182	9.910e+01	196	7.930e+00
185	2.318e+02	197	4.441e-16
186	1.228e+02	198	1.081e+00
189	1.063e+01	200	9.326e-15
190	4.545e+00	201	9.770e-15
191	2.337e-01	205	1.221e-15
202	3.558e+00	206	8.559e+00
204	5.314e-02	209	7.930e+00
207	1.000e-01	211	1.081e+00
208	4.489e-01	213	7.883e-15
210	3.591e-01	214	9.770e-15
212	2.127e-01	216	1.531e-01
215	6.880e-01	217	1.400e-01
218	8.190e-01	219	1.750e-01
229	7.573e+00	220	8.711e+00
231	1.589e-01	221	9.885e-01
232	2.528e-01	222	2.734e-01
244	1.640e-02	223	8.083e+00
248	3.018e+00	225	1.256e+00
250	2.414e+00	226	1.531e-01
253	3.182e-01	227	1.400e-01
259	1.720e-01	228	1.750e-01
260	5.823e+01	233	8.559e+00
264	9.182e+00	234	7.884e-01
271	9.818e+00	236	7.930e+00
272	2.650e+01	238	1.081e+00
273	4.450e+01	240	6.106e-15
274	1.446e+01	241	7.994e-15
275	2.430e+01	246	8.559e+00
276	5.241e+01	247	9.467e+00
277	2.009e+01	249	7.930e+00
282	3.292e+01	251	1.081e+00
285	6.447e+00	255	8.607e-02
286	1.457e+00	256	1.535e+00
288	1.000e-01	257	3.013e+01
290	6.400e-01	258	4.606e-01
293	1.400e+00	261	1.786e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
296	4.500e-01	262	3.777e-01
299	1.500e-01	263	6.929e-02
301	1.031e+01	268	1.786e-01
313	1.643e+02	269	3.777e-01
314	6.870e+01	270	6.929e-02
315	1.296e+02	279	2.158e-01
316	5.418e+01	281	6.192e-03
317	2.000e+02	283	1.388e-17
318	1.750e+02	284	6.192e-03
321	1.250e+02	320	9.710e-02
323	3.750e+02	322	1.846e-01
324	1.250e+02	329	2.906e-01
325	2.849e+02	330	9.691e-01
328	9.788e+01	337	2.906e-01
331	2.849e+02	338	1.780e+01
332	9.788e+01	344	2.762e-03
333	2.499e+02	346	2.713e-03
336	9.098e+01	347	1.006e-04
339	2.499e+02	356	3.000e-03
340	9.098e+01	357	2.430e-02
341	6.064e+00	360	5.421e-20
342	6.064e+00	364	1.693e+00
343	5.909e+01	365	4.441e-16
348	1.469e+01	366	1.290e+00
349	3.219e+02	367	4.305e-01
350	1.503e+02	371	6.688e-01
351	7.752e+00	373	4.441e-16
352	3.219e+02	377	1.693e+00
355	1.580e+02	378	4.163e-16
358	3.542e+02	379	1.290e+00
359	1.834e+02	384	6.688e-01
361	2.637e+01	385	1.665e-16
362	7.304e+00	386	4.996e-16
368	4.186e+00	388	5.577e-01
370	3.349e+00	389	5.577e-01
372	3.225e-01	390	2.250e+00
375	3.103e+00	391	5.577e-01
376	4.544e-01	392	1.847e+00
380	2.250e-01	393	8.815e-01
387	6.231e-02	394	1.004e+00
401	6.020e-01	395	5.577e-01
403	1.492e+00	397	1.227e+00
415	1.034e+01	398	5.577e-01
416	2.268e+00	399	5.577e-01
422	3.428e-02	400	5.577e-01
434	2.255e-01	402	5.551e-16
436	1.804e-01	404	1.693e+00
440	3.769e-02	405	1.665e-15
441	4.406e+00	406	1.290e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
442	7.854e-01	407	5.188e-01
444	2.643e-01	408	1.665e-15
452	7.727e-01	411	6.688e-01
460	2.464e-01	412	1.221e-15
462	1.971e-01	413	1.665e-15
467	8.872e-02	414	1.221e-15
469	1.697e+00	417	1.693e+00
474	1.798e-01	418	1.665e-15
478	6.171e+01	419	1.290e+00
482	7.674e-01	420	4.783e-01
483	2.650e+01	423	2.331e-15
492	2.723e+01	424	6.688e-01
494	4.172e+01	425	1.221e-15
498	4.664e+00	426	1.665e-15
507	1.331e-01	427	1.221e-15
508	1.335e+01	430	1.693e+00
509	1.950e+01	431	3.331e-16
510	5.417e+01	432	1.290e+00
511	2.191e+01	433	4.599e-01
514	1.000e-01	437	6.688e-01
526	4.062e+01	439	2.220e-16
528	4.029e+00	443	1.693e+00
529	2.000e+01	445	1.290e+00
547	1.574e+00	446	1.886e+00
549	1.500e-01	450	6.688e-01
550	4.560e-01	456	1.693e+00
551	1.500e-01	457	6.661e-16
552	3.000e-02	458	1.290e+00
553	1.899e-02	459	2.944e-01
555	9.000e+00	463	6.688e-01
556	1.140e+00	464	2.220e-16
558	5.895e-01	465	6.661e-16
563	4.139e+00	466	2.220e-16
564	3.583e-01	470	1.873e-01
565	2.958e-01	471	1.107e+00
567	1.457e+00	472	2.125e+01
569	1.000e-01	473	4.366e-01
571	6.400e-01	475	8.395e-01
574	1.400e+00	476	1.088e+00
577	4.500e-01	477	7.451e-01
580	1.500e-01	479	9.930e-02
582	1.031e+01	480	2.165e-01
594	2.523e+02	481	4.913e-02
595	8.794e+01	485	8.395e-01
596	1.989e+02	486	1.088e+00
597	6.935e+01	487	7.451e-01
598	2.000e+02	489	9.930e-02
599	3.000e+02	490	2.165e-01
602	1.250e+02	491	4.913e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
604	5.000e+02	495	1.041e+00
605	1.250e+02	496	1.289e+00
606	3.926e+02	497	9.462e-01
609	1.077e+02	499	4.136e-01
612	3.926e+02	500	3.232e-01
613	1.077e+02	501	1.558e-01
614	3.151e+02	502	1.067e-01
617	6.521e+01	503	1.067e-01
620	3.151e+02	504	9.200e-02
621	6.521e+01	505	9.443e-02
622	1.710e+01	506	3.432e-01
623	1.104e+01	513	1.180e-01
624	7.045e+01	515	3.710e-01
625	2.308e+01	516	6.197e-01
632	2.984e+01	517	2.766e-01
634	5.798e+02	518	3.627e-01
635	2.405e+02	519	3.627e-01
636	1.736e+01	520	3.590e-01
637	5.798e+02	521	3.590e-01
640	2.578e+02	522	3.627e-01
643	5.681e+02	523	4.770e-01
644	2.920e+02	524	3.590e-01
646	5.904e+01	525	4.163e-17
647	6.877e+00	530	5.612e+00
648	1.021e+01	531	5.612e+00
649	2.232e+00	532	8.223e+00
664	2.680e-02	533	1.808e+00
667	2.389e-01	534	2.341e+00
669	1.911e-01	535	1.664e+00
674	1.671e+00	536	1.496e+00
675	5.864e-01	537	1.572e+00
679	3.500e-01	538	1.447e+00
686	3.500e-01	539	1.271e+00
700	5.598e-01	601	9.710e-02
702	1.216e+00	603	5.705e-02
714	9.960e+00	610	9.820e-02
715	1.168e+00	611	4.944e-01
720	4.373e+00	618	9.820e-02
721	7.675e-02	619	1.072e+01
722	3.498e+00	626	2.460e-03
724	5.161e-01	627	2.460e-03
730	4.436e-01	630	6.846e-04
746	6.161e-01	631	7.563e-05
748	4.929e-01	633	4.770e-03
753	5.519e+00	641	9.000e-04
754	3.432e-01	642	1.720e-02
764	8.636e-01	650	1.096e+00
772	3.080e-01	652	8.402e-01
774	2.464e-01	653	2.967e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
779	9.759e-01	657	4.312e-01
781	1.933e+00	659	1.110e-16
786	3.200e-01	661	1.110e-16
794	3.042e+01	663	1.096e+00
795	2.519e+01	665	8.402e-01
796	4.227e+01	666	3.146e-01
800	4.246e+01	670	4.312e-01
805	5.813e+00	672	5.551e-17
810	8.489e+00	676	1.096e+00
820	1.071e+01	677	8.327e-17
821	1.855e+01	678	8.402e-01
822	5.687e+01	680	6.384e-16
823	2.355e+01	683	4.312e-01
824	1.000e-01	684	1.943e-16
826	3.500e-01	687	4.406e-01
829	1.730e-01	688	4.406e-01
838	4.375e+01	689	1.536e+00
840	3.500e+01	690	4.406e-01
841	2.500e+01	691	1.281e+00
853	1.545e-01	692	7.220e-01
856	1.070e-01	693	7.670e-01
859	2.351e+00	694	4.406e-01
860	1.500e-01	695	3.273e-02
861	2.250e-01	696	8.718e-01
862	4.685e-01	697	4.406e-01
863	3.750e-02	698	4.406e-01
864	3.000e-01	699	4.406e-01
865	3.424e-01	701	3.886e-16
867	3.553e+00	703	1.096e+00
868	9.348e-01	704	2.776e-16
870	6.779e-01	705	8.402e-01
875	1.795e+01	706	3.503e-01
876	2.329e-01	707	6.106e-16
877	8.873e-02	708	5.551e-17
886	1.574e+00	710	4.312e-01
888	1.500e-01	711	2.776e-16
889	4.560e-01	712	3.886e-16
890	1.500e-01	713	2.776e-16
891	3.000e-02	716	1.096e+00
892	1.899e-02	717	2.220e-16
894	9.000e+00	718	8.402e-01
895	1.140e+00	719	3.478e-01
897	5.895e-01	723	4.312e-01
902	4.139e+00	725	2.776e-16
903	3.583e-01	726	5.551e-17
904	2.958e-01	727	3.461e-02
906	1.457e+00	728	1.110e-16
908	1.000e-01	729	1.096e+00
910	6.400e-01	731	8.402e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
913	1.400e+00	732	3.235e-01
916	4.500e-01	733	2.220e-16
919	1.500e-01	736	4.312e-01
921	1.031e+01	737	1.110e-16
933	3.648e+02	738	1.110e-16
934	1.126e+02	740	5.551e-17
935	2.860e+02	742	1.096e+00
936	8.713e+01	744	8.402e-01
937	2.000e+02	745	3.775e-01
938	4.000e+02	749	4.312e-01
941	1.000e+02	755	1.096e+00
943	6.000e+02	756	5.551e-17
944	1.000e+02	757	8.402e-01
945	4.000e+02	758	1.236e+00
946	1.002e+02	759	5.551e-17
948	7.425e+00	762	4.312e-01
949	1.002e+02	763	5.551e-17
951	5.002e+02	765	5.551e-17
952	1.077e+02	766	1.388e-16
953	3.151e+02	768	1.096e+00
954	6.030e+01	770	8.402e-01
957	6.030e+01	771	2.142e-01
959	3.754e+02	775	4.312e-01
960	6.030e+01	782	7.628e-02
961	3.719e+01	783	6.230e-01
962	2.009e+01	784	1.283e+01
963	8.314e+01	785	2.213e-01
964	8.299e+01	787	7.345e-01
967	2.263e+01	788	1.017e+00
971	7.461e+01	789	6.843e-01
973	1.009e+03	791	1.023e-01
974	3.862e+02	792	1.692e-01
975	4.339e+01	793	6.220e-02
976	6.329e+02	797	7.345e-01
977	3.765e+02	798	1.017e+00
979	5.309e+01	799	6.843e-01
980	3.765e+02	801	1.023e-01
982	9.247e+02	802	1.692e-01
983	4.561e+02	803	6.220e-02
985	1.476e+02	804	1.388e-17
986	1.719e+01	807	8.132e-01
988	9.696e+00	808	7.841e-01
994	2.144e+00	809	4.509e-01
1000	6.000e-01	811	2.757e-01
1007	4.251e-01	812	2.130e-01
1010	3.058e-01	813	1.060e-01
1019	5.250e-01	814	4.470e-02
1040	5.276e-01	815	4.380e-02
1054	1.628e+01	816	3.300e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1055	4.633e+00	817	2.083e-01
1060	4.686e+00	818	4.913e-01
1061	1.919e-01	819	1.581e-01
1062	3.749e+00	825	4.763e-02
1064	8.285e-01	827	5.021e-02
1073	1.032e+00	828	3.332e-01
1076	7.423e-01	830	7.205e-02
1081	9.809e-01	831	7.205e-02
1083	6.773e-01	832	6.985e-02
1086	7.885e-01	833	6.985e-02
1089	3.257e-01	834	7.205e-02
1093	2.937e+00	835	1.175e-01
1099	1.881e+00	836	6.985e-02
1101	1.492e+00	837	1.388e-17
1104	9.545e-01	839	3.469e-18
1112	4.031e-01	842	2.857e+00
1115	2.900e-01	843	2.857e+00
1119	2.803e+00	844	4.580e+00
1120	6.129e-01	845	9.056e-01
1121	2.202e+00	846	1.254e+00
1126	4.608e-01	847	8.218e-01
1134	3.326e+01	848	7.148e-01
1135	2.205e+01	849	7.309e-01
1136	4.257e+01	850	6.526e-01
1140	3.678e+01	851	5.329e-01
1145	1.420e+01	879	5.484e+00
1150	1.545e+01	881	4.801e+00
1160	1.431e+01	883	2.349e+00
1161	1.555e+01	884	5.908e+00
1162	3.841e+01	929	9.860e-02
1163	2.234e+01	940	9.710e-02
1164	3.000e-01	942	4.047e-03
1169	2.249e-01	950	2.250e-01
1172	1.116e+01	958	6.442e+00
1176	6.000e-01	965	6.206e-04
1178	4.729e+01	966	6.206e-04
1180	4.625e+01	969	9.793e-04
1181	3.250e+01	970	2.157e-05
1192	9.200e-01	972	8.964e-04
1193	9.269e-02	981	1.190e-02
1196	1.314e-01	987	8.964e-04
1199	3.744e+00	990	7.298e-01
1200	4.500e-01	992	2.768e-01
1201	5.625e-01	993	8.207e-02
1202	4.820e-01	1001	8.327e-16
1203	4.688e-02	1003	7.298e-01
1204	8.250e-01	1005	2.768e-01
1205	4.638e-01	1006	9.287e-02
1207	6.920e+00	1008	3.053e-16

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1208	1.701e+00	1009	1.665e-15
1210	1.457e+00	1011	6.384e-16
1215	1.546e+01	1012	2.498e-16
1216	3.842e-01	1013	1.860e-15
1217	1.154e-01	1014	1.521e-01
1220	1.545e-01	1015	1.306e-01
1223	1.070e-01	1016	6.305e-01
1226	2.351e+00	1017	1.881e-01
1227	1.500e-01	1018	4.265e-01
1228	2.250e-01	1020	2.701e-01
1229	4.685e-01	1021	1.521e-01
1230	3.750e-02	1023	1.881e-01
1231	3.000e-01	1024	1.521e-01
1232	3.424e-01	1025	1.306e-01
1234	3.553e+00	1026	1.881e-01
1235	9.348e-01	1027	5.818e-01
1237	6.779e-01	1028	5.603e-01
1242	1.795e+01	1029	1.060e+00
1243	2.329e-01	1030	6.178e-01
1244	8.873e-02	1031	8.562e-01
1253	1.574e+00	1032	6.591e-01
1255	1.500e-01	1034	5.818e-01
1256	4.560e-01	1035	1.304e+00
1257	1.500e-01	1036	6.178e-01
1258	3.000e-02	1037	5.818e-01
1259	1.899e-02	1038	5.603e-01
1261	9.000e+00	1039	6.178e-01
1262	1.140e+00	1041	2.490e-01
1264	5.895e-01	1042	2.274e-01
1269	4.139e+00	1043	7.273e-01
1270	3.583e-01	1044	2.849e-01
1271	2.958e-01	1045	5.234e-01
1273	1.457e+00	1046	3.311e-01
1275	1.000e-01	1047	4.444e-01
1277	6.400e-01	1048	2.490e-01
1280	1.400e+00	1050	2.849e-01
1283	4.500e-01	1051	2.490e-01
1286	1.500e-01	1052	2.274e-01
1288	1.031e+01	1053	2.849e-01
1300	5.089e+02	1056	7.298e-01
1301	1.441e+02	1058	2.768e-01
1302	3.865e+02	1059	1.280e-01
1303	1.005e+02	1065	2.776e-16
1304	2.000e+02	1067	2.561e-02
1305	4.000e+02	1069	7.298e-01
1306	8.271e+01	1071	2.768e-01
1309	8.271e+01	1072	9.837e-02
1310	6.827e+02	1074	1.943e-16
1311	8.271e+01	1075	1.610e-15

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1312	4.000e+02	1077	6.384e-16
1313	2.079e+02	1078	2.498e-16
1316	1.077e+02	1079	1.860e-15
1318	6.079e+02	1080	8.327e-16
1319	1.077e+02	1082	7.298e-01
1320	3.151e+02	1084	2.768e-01
1321	7.815e+01	1085	1.779e-01
1324	1.785e+01	1087	2.498e-16
1326	3.932e+02	1088	1.665e-15
1327	1.785e+01	1090	6.661e-16
1328	7.374e+01	1091	1.943e-16
1329	3.656e+01	1092	1.915e-15
1331	3.094e+02	1095	7.298e-01
1333	2.263e+01	1097	2.768e-01
1338	1.865e+02	1098	3.700e-01
1340	1.595e+03	1105	1.527e-16
1341	5.196e+02	1106	7.772e-16
1342	6.557e+01	1108	7.298e-01
1343	1.218e+03	1110	2.768e-01
1344	3.765e+02	1111	3.207e-02
1346	5.852e+02	1113	2.498e-16
1349	1.518e+03	1114	1.554e-15
1350	3.263e+02	1116	5.829e-16
1352	2.230e+02	1117	1.388e-16
1353	4.298e+01	1118	1.804e-15
1355	1.099e+01	1122	1.240e-02
1361	1.484e+00	1123	3.337e-01
1374	6.802e-01	1124	7.927e+00
1377	4.893e-01	1125	9.490e-02
1386	7.000e-01	1127	6.544e-01
1407	4.416e-01	1128	7.489e-01
1421	1.603e+01	1129	5.550e-01
1422	5.070e+00	1130	9.714e-17
1427	1.309e+01	1131	8.600e-02
1428	2.899e-01	1132	1.274e-01
1429	6.247e+00	1133	5.759e-02
1430	3.240e+00	1137	6.544e-01
1431	1.107e+00	1138	7.489e-01
1432	1.045e+00	1139	5.550e-01
1433	9.500e-01	1141	8.600e-02
1440	4.128e+00	1142	1.274e-01
1443	2.969e+00	1143	5.759e-02
1450	9.754e-01	1144	1.388e-17
1453	4.031e+00	1147	4.875e-01
1456	1.924e+00	1148	7.714e-01
1466	4.563e+00	1149	5.774e-01
1469	3.282e+00	1151	1.804e-01

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores λ 0 do problema BNL2

Coordenada y	Valor y (Dual)
7	8.830e-01
9	2.854e+00
12	2.651e+01
13	3.125e+01
18	1.993e+00
20	2.965e-01
22	3.073e-01
24	1.405e-01
25	4.335e-02
27	1.852e+00
29	3.890e-02
30	3.890e-02
42	1.924e+00
50	5.626e+00
51	3.365e+00
52	7.835e+00
53	7.741e+00
57	5.012e+00
58	3.307e+00
59	1.445e+01
60	1.615e+01
61	1.855e+01
62	1.237e+01
63	2.117e+00
78	7.835e+00
106	1.435e+01
107	1.435e+01
108	1.289e+01
109	1.428e+01
110	1.435e+01
111	1.435e+01
112	1.347e+01
114	1.194e+00
117	1.077e+01
118	1.402e+01
123	8.649e-01
128	2.708e-01
129	9.319e-03
134	1.405e-01
135	4.335e-02
137	8.374e-01
139	3.890e-02
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Coordenada y	Valor y (Dual)
140	3.890e-02
154	1.044e+00
161	2.771e+00
162	6.296e+00
163	1.310e+01
164	9.390e+00
168	2.469e+00
169	1.515e+00
170	7.092e+00
171	8.123e+00
172	9.579e+00
173	4.797e+00
174	9.697e-01
189	1.310e+01
231	6.801e-01
238	9.577e-01
239	1.495e+00
243	7.144e-01
244	1.278e+00
248	5.825e+00
249	5.649e+00
250	7.852e+00
251	7.902e+00
252	7.701e+00
286	4.291e-01
291	1.725e-01
292	8.237e-03
297	1.405e-01
298	4.335e-02
300	4.294e-01
302	3.890e-02
303	3.890e-02
309	6.334e+00
321	5.464e-01
323	7.609e-03
329	1.504e+00
330	8.483e-01
331	2.015e+00
332	2.012e+00
336	1.340e+00
337	7.839e-01
338	3.842e+00
339	4.468e+00
340	5.351e+00
341	2.416e+00
342	5.017e-01
357	2.015e+00
399	4.347e-01
400	6.158e-01
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Coordenada y	Valor y (Dual)
405	1.005e+00
413	1.237e+00
414	8.334e-01
418	1.932e+00
419	1.738e+00
420	7.555e-01
424	3.777e+00
425	4.341e+00
426	3.843e+00
427	5.076e+00
428	5.107e+00
429	4.985e+00
490	1.776e-01
495	8.467e-03
496	2.804e-01
499	2.387e-03
500	8.495e-02
501	4.992e-03
506	1.405e-01
507	4.335e-02
509	2.821e-01
511	3.890e-02
512	3.890e-02
517	3.611e+00
519	4.838e+00
520	5.555e+00
521	1.446e+01
522	1.485e+00
524	1.535e+00
525	2.000e-03
526	1.250e+00
538	3.914e-01
540	4.847e-03
546	9.612e-01
547	5.722e-01
548	1.336e+00
549	1.313e+00
553	8.564e-01
554	5.128e-01
555	2.617e+00
556	3.009e+00
557	3.607e+00
558	1.038e+00
559	3.282e-01
574	1.336e+00
604	1.912e+00
605	3.416e+00
607	3.359e-01
608	3.863e-01
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Coordenada y	Valor y (Dual)
610	3.463e-01
611	5.427e-01
612	1.177e-01
613	1.177e-01
616	2.002e-01
618	1.177e-01
619	1.177e-01
620	1.177e-01
621	1.177e-01
622	1.177e-01
623	1.177e-01
624	5.530e-01
625	4.535e-01
626	4.922e-02
629	4.636e-01
630	3.545e-01
635	1.137e+00
636	1.728e+00
637	1.103e+00
638	2.068e+00
639	2.087e+00
640	2.013e+00
728	8.660e-02
733	3.697e-03
734	1.224e-01
738	1.766e-03
739	6.286e-02
740	2.553e-03
745	1.405e-01
746	4.335e-02
748	1.909e-01
750	3.890e-02
751	3.890e-02
756	3.834e+00
758	2.652e+00
759	2.970e+00
760	4.582e+00
761	1.002e+00
763	6.412e-01
764	1.939e-03
765	7.542e-01
777	2.789e-01
779	2.387e-03
785	6.272e-01
786	2.017e-01
787	6.363e-01
788	6.062e-01
792	5.588e-01
793	3.449e-01
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Coordenada y	Valor y (Dual)
794	1.806e+00
795	2.048e+00
796	2.454e+00
797	5.208e-01
798	2.207e-01
799	1.504e-01
813	6.363e-01
816	2.279e-01
841	1.163e+00
842	2.800e+00
843	1.573e+00
844	1.229e-02
845	2.708e-01
846	2.297e-01
848	1.654e-01
849	3.367e-01
850	7.242e-02
851	7.242e-02
852	3.190e-02
854	1.120e-02
856	7.242e-02
857	7.242e-02
858	7.242e-02
859	7.242e-02
860	7.242e-02
861	7.242e-02
862	2.508e-01
863	2.094e+00
864	1.615e-01
865	4.217e-02
868	3.043e-01
869	2.377e-01
874	6.477e-01
875	8.509e-01
876	6.845e-01
877	1.277e+00
878	1.288e+00
879	1.243e+00
995	5.250e-02
1000	2.800e-03
1001	9.272e-02
1005	1.766e-03
1006	3.127e-02
1007	1.857e-03
1008	8.877e-04
1012	1.405e-01
1013	4.335e-02
1015	1.182e-01
1017	3.890e-02
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Coordenada y	Valor y (Dual)
1018	3.890e-02
1023	3.024e+00
1025	1.333e+00
1026	1.420e+00
1027	1.354e+00
1028	4.684e-01
1030	1.513e-01
1031	1.297e-03
1032	3.543e-01
1044	1.808e-01
1046	1.766e-03
1052	3.860e-01
1053	1.200e-01
1054	3.848e-01
1055	3.695e-01
1059	3.439e-01
1060	2.135e-01
1061	1.177e+00
1062	1.325e+00
1063	1.600e+00
1064	3.158e-01
1065	1.366e-01
1066	9.406e-02
1080	3.848e-01
1083	1.405e-01
1109	4.944e-01
1110	7.032e-01
1111	5.227e-01
1113	3.924e-02
1114	2.418e-02
1116	9.969e-02
1117	2.648e-01
1118	4.174e-02
1119	2.648e-02
1121	2.234e+00
1124	4.174e-02
1125	4.174e-02
1126	4.174e-02
1127	4.174e-02
1128	4.174e-02
1129	4.174e-02
1130	1.697e-01
1131	1.018e+00
1133	2.304e-02
1142	2.026e-01
1143	3.442e-01
1144	2.103e-01
1145	5.909e-01
1146	5.977e-01

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Coordenada y	Valor y (Dual)
1147	5.705e-01
1290	3.190e-02
1295	1.264e-03
1296	4.184e-02
1300	5.103e-04
1301	1.816e-02
1302	9.345e-04
1307	1.405e-01
1308	4.335e-02
1310	7.170e-02
1312	3.890e-02
1313	3.890e-02
1318	2.232e+00
1320	4.241e-01
1321	4.321e-01
1322	4.061e-01
1323	1.682e-01
1324	9.807e-01
1327	8.942e-04
1328	1.229e-01
1338	6.853e-03
1340	1.099e-01
1342	1.766e-03
1348	1.925e-01
1349	6.126e-02
1350	1.952e-01
1351	1.848e-01
1355	2.076e-01
1356	1.295e-01
1357	7.142e-01
1358	8.037e-01
1359	9.707e-01
1360	1.919e-01
1361	8.287e-02
1362	8.941e-02
1376	1.952e-01
1379	1.175e-01
1404	9.762e-02
1405	2.249e-01
1406	1.091e-01
1411	5.926e-02
1412	2.041e-01
1422	1.000e-04
1423	1.000e-04
1425	1.000e-04
1426	5.230e-02
1428	5.926e-02
1429	2.041e-01
1430	6.167e-04
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Coordenada y	Valor y (Dual)
1434	1.538e-04
1441	4.550e-03
1585	1.322e-02
1587	5.667e-03
1590	7.056e-04
1591	2.336e-02
1596	7.107e-03
1599	6.370e-05
1602	1.405e-01
1603	4.335e-02
1605	5.010e-02
1607	3.890e-02
1608	3.890e-02
1616	2.957e-03
1619	4.470e-02
1622	6.167e-04
1633	1.496e-03
1635	8.428e-02
1637	5.103e-04
1643	2.336e-02
1644	8.415e-03
1645	2.440e-02
1646	2.388e-02
1650	1.275e-01
1651	7.985e-02
1652	5.300e-01
1653	5.845e-01
1654	6.855e-01
1655	1.139e-01
1656	5.111e-02
1657	1.619e-01
1671	2.440e-02
1674	1.511e-01
1721	1.000e-04
1880	1.095e-02
1885	2.940e-04
1891	2.966e-03
1894	2.658e-05
1897	1.405e-01
1898	4.335e-02
1900	2.367e-02
1902	3.890e-02
1903	3.890e-02
1930	6.255e-02
1938	8.954e-03
1939	6.526e-03
1940	1.422e-02
1941	1.393e-02
1945	1.360e-02
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Coordenada y	Valor y (Dual)
1946	4.286e-02
1947	3.845e-01
1948	4.175e-01
1949	4.785e-01
1950	6.660e-02
1951	2.743e-02
1952	1.231e-01
1966	1.422e-02
1969	7.036e-02
2175	7.932e-03
2191	4.595e-02
2196	3.716e-04
2197	1.185e-04
2201	1.405e-01
2202	4.335e-02
2204	3.563e-02
2207	3.890e-02
2208	3.890e-02
2210	7.460e-02
2234	4.570e-03
2236	4.033e-02
2256	5.406e-03
2257	3.934e-03
2258	8.577e-03
2259	8.398e-03
2263	8.200e-03
2264	2.906e-02
2265	2.463e-01
2266	2.663e-01
2267	3.033e-01
2268	4.681e-02
2269	1.860e-02
2270	8.077e-02
2284	8.577e-03
2287	9.682e-02